The evolution of 3-points shooting in NBA and its importance in modern basketball

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Problem Statement

Since the introduction of the 3-point line in the 1979–80 season, the role of the 3-point shot in the NBA has evolved dramatically. In recent years, teams have increasingly relied on 3-point shooting as a core offensive strategy, with players such as Stephen Curry helping accelerate this trend.

However, the extent and timing of this strategic shift are not always clearly quantified. Understanding **how three-point shooting has evolved over time and how teams and players have adapted their strategies** is essential for evaluating modern offensive trends.

This analysis aims to explore the historical evolution of 3-point shooting in the NBA since its introduction, focusing on changes in **shot volume, efficiency, team strategy, and player behaviour**.

The goal is to identify **key trends and strategic shifts** that have contributed to the modern 3-point focused style of play.

Business Impact

Understanding the evolution of 3-point shooting provides valuable insights for basketball decision-makers, including coaching staff, front offices, and analysts.

Key impacts of this analysis include:

1. **Offensive Strategy Optimization**

By identifying how successful teams have increased their reliance on 3-point shooting, teams can better evaluate whether their offensive strategy aligns with modern trends in the NBA.

2. **Player Recruitment and Development**

Insights into the relationship between **3-point volume and efficiency** can help teams prioritize players who provide the greatest offensive value in today's game.

3. **Competitive Benchmarking**

Teams can compare their shot selection strategies with league trends to determine whether they are **keeping pace with modern offensive strategies**.

4. **Game Strategy and Analytics**

Understanding how the 3-point shot affects scoring efficiency can support **data-driven coaching decisions**, such as lineup construction, shot selection, and offensive schemes.

5. **Long-Term Strategic Planning**

Front offices can use historical trends to evaluate how the game is evolving and anticipate future shifts in offensive play.

General Dataset Information

Files name: Player Totals.csv

Team Totals.csv

Description: Basketball statistics from 1947 to present season.

Datasets details: 33120 Rows & 33 Columns

1908 Rows & 28 Columns

Size: 4,388 KB (4.28 MB)

273 KB

Source: Kaggle – Dataset Link

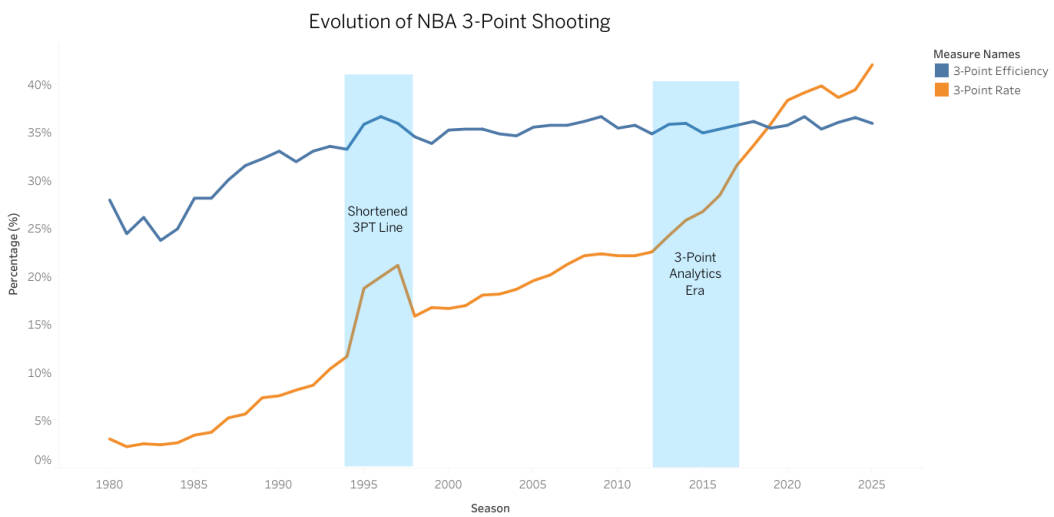
Target Features

The columns that will be used to simplify and analyse the problem statement are:

- season: range of seasons, from 1980 to 2025.
- team: name of teams that have been part of the league during this period of time.
- player: player's names that have been part of the league.
- g: number of games during the season.
- pos: Position in game of the player.
- mp: Minutes played in a season by player.
- x3p: 3-Points made during season by teams and players.
- x3pa: 3-Points attempts during season by teams and players.
- fg: Field Goals made during season by teams and players.
- fga: Field Goals attempts made during season by teams and players.

List of Analysis

1. **League 3-Point Rate and Efficiency over time.**



Since the introduction of the three-point line in the **1979–80 NBA season**, the use of 3-point shots has increased steadily across the league. However, the rate of adoption varied over time, creating distinct eras in the evolution of 3-point shooting.

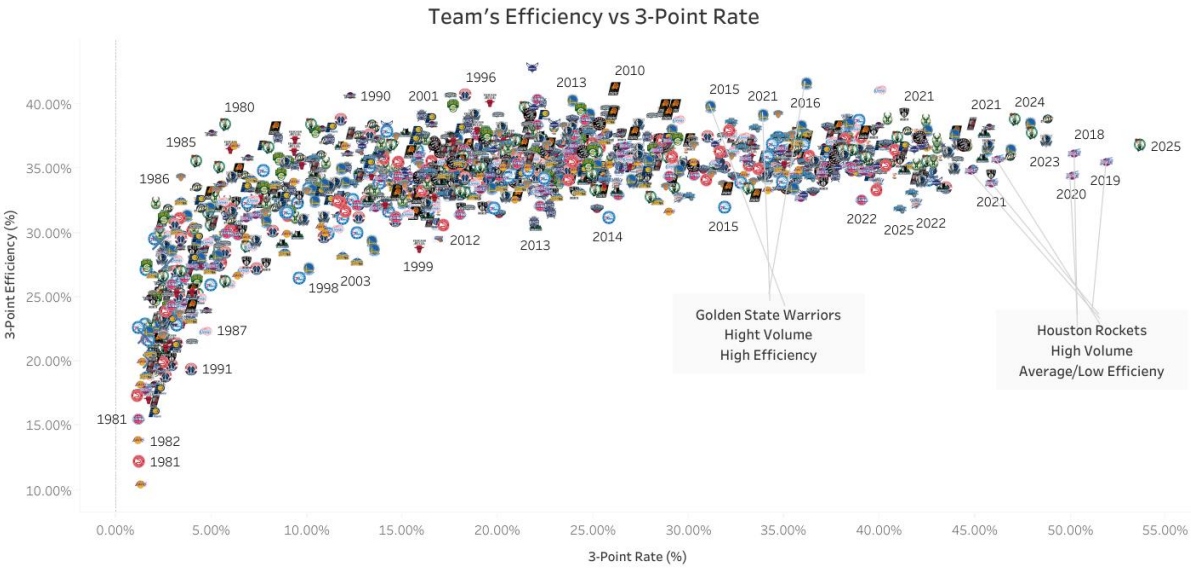
Assimilation Era (1980–early 1990's)
Following its introduction, teams were slow to incorporate the 3-point shot into their offensive strategies. During the first decade, the share of 3-point attempts increased gradually from roughly **3% to about 12% of total Field Goal Attempts**.

Shortened Line Distance (1994–1997)
To encourage perimeter shooting, the NBA temporarily shortened the 3-point distance starting in the **1993–94 season**. This change led to a noticeable spike in 3-point attempts. After the league restored the original distance, usage declined slightly but continued to grow steadily throughout the 2000's.

3-Point Revolution (mid-2010's onward)
The rise of perimeter-focused offenses accelerated in the 2010's, influenced heavily by players like Stephen Curry and Klay Thompson with the Golden State Warriors. Beginning around the **2015–16 season**, the league experienced a sharp increase in 3-point attempts as teams adopted high-volume three-point strategies.

Interestingly, this dramatic rise in shot volume contrasts with **3-point efficiency**, which has remained relatively stable since the early years of adoption, fluctuating mostly between **32% and 37%**.

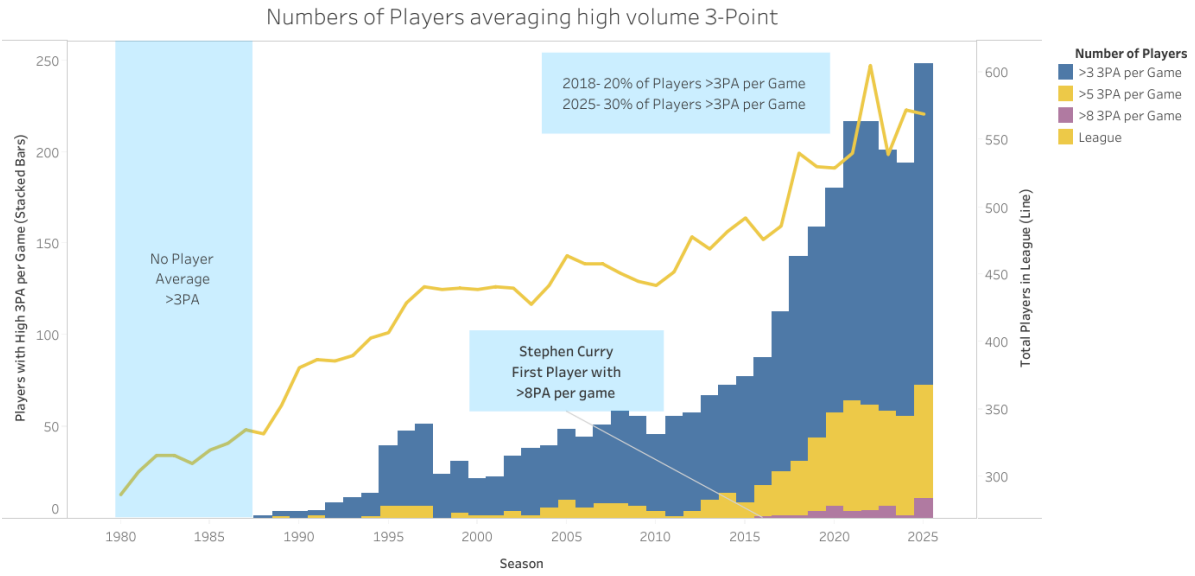
2. **Attempts vs Efficiency by Teams.**



This analysis is consistent with previous observations: in the early years, teams had both **low 3-point usage and low efficiency**. By the mid-1990s, **efficiency stabilized**, with most teams clustered between **32–37%**, while the **volume of 3-point attempts increased** across the league. Notably, some teams emerged as **pioneers of the 3-point strategy**, combining **high volume with high efficiency**, setting the stage for modern perimeter-oriented offenses.

In recent years, the NBA has seen an unprecedented volume of 3-point attempts, with many teams adopting a high-paced, perimeter-focused strategy. Despite the increase in attempts, league-wide 3-point efficiency has plateaued, indicating a balance between volume and accuracy in the modern game.

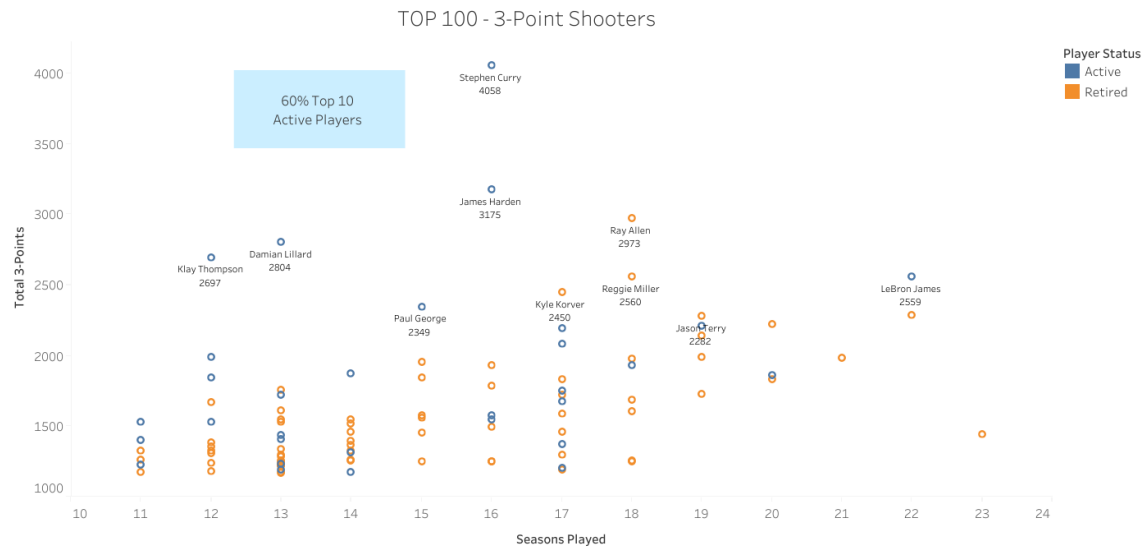
3. **Number of players attempting 3 / 5 / 8 3-Point Shots per game.**



During the **Assimilation Era**, it was rare for any player to average **more than 3 attempts per game** from beyond the 3-point line. The number of players averaging more than 3 attempts increased considerably during the **Shortened Line** era and continued to grow gradually throughout the 2000's, with a few players beginning to **exceed 5 attempts per game**. The **2015–16** season can be considered a turning point: for the first time a player averaged more than **8 three-point attempts per game** (Stephen Curry). Since then, an increasing number of players have joined the group of high-volume 3-point shooters.

This shift reflects the broader transformation of offensive strategies in the modern NBA, where 3-point shooting has become a central component of team offenses.

4. **TOP 100 3-Point Shooters.**



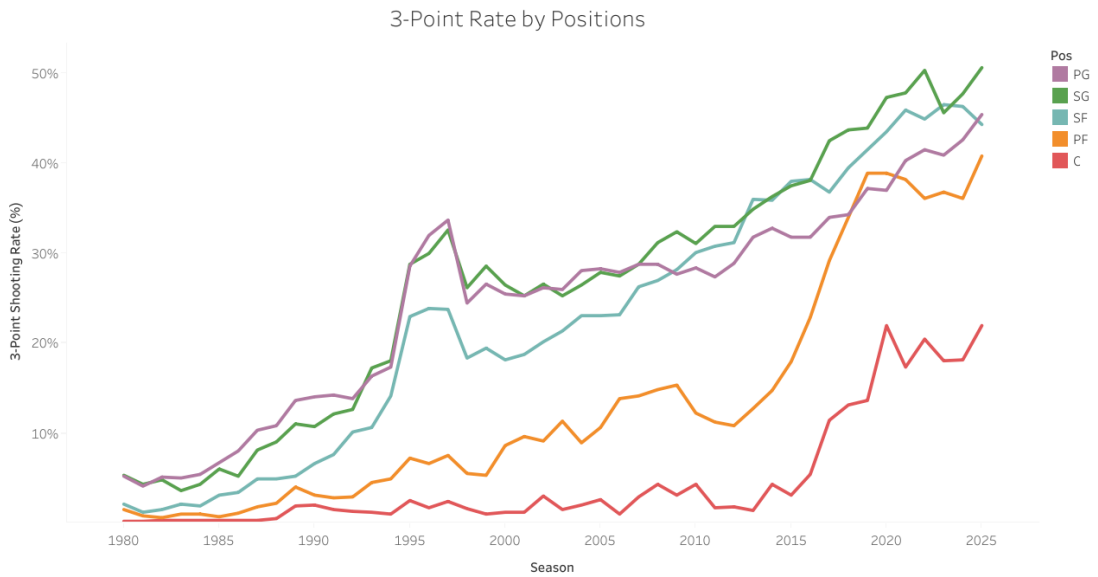
Although only **31% of the Top 100 players in total 3-point field goals are currently active**, these players are disproportionately represented at the top of the leaderboard. Active players account for **21% of the Top 50** and dominate the **Top 10 with 60% of the players still active**.

This pattern reflects the increasing importance of the three-point shot in modern NBA offenses. The concentration of active players among the highest ranks suggests that current players accumulate three-point totals at a much faster pace than players from previous eras.

This trend is largely driven by the modern offensive strategies adopted across the league, where three-point shooting has become a central component of team scoring systems. Players such as Stephen Curry, James Harden, and Damian Lillard exemplify this shift, as they have reached historically high totals while still active.

This distribution suggests that the leaderboard is likely to continue changing rapidly as active players climb the rankings in a league increasingly defined by high-volume three-point shooting.

5. **3-Point Shooting Rate by Position.**



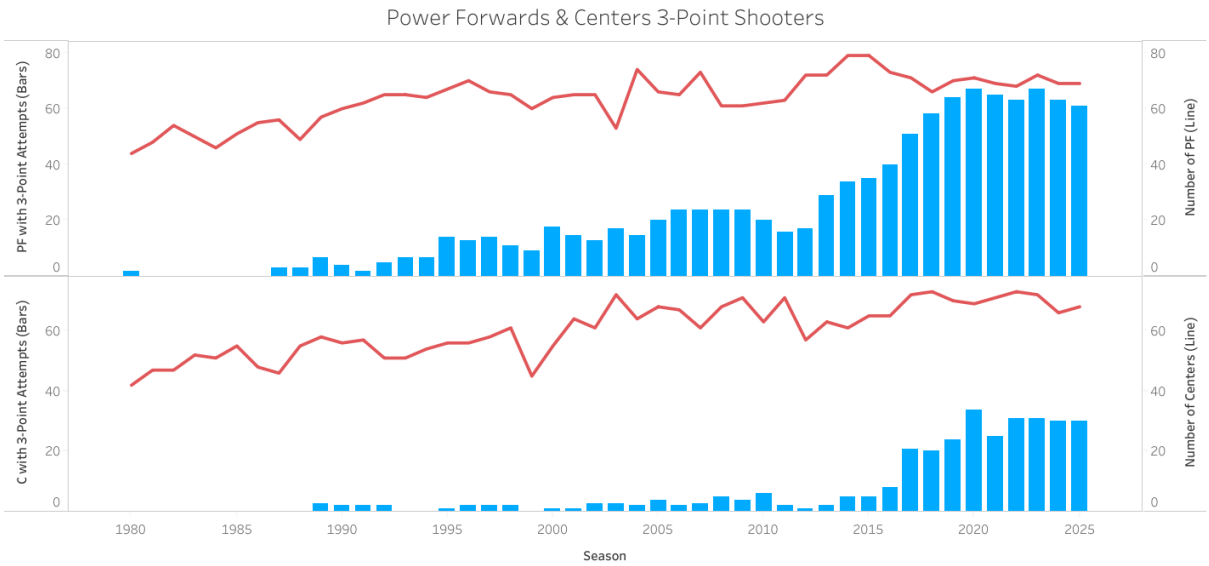
Traditionally, 3-point shooting has been associated with the **perimeter positions** in basketball. Point Guards (PG), Shooting Guards (SG), and Small Forwards (SF) have historically taken the majority of shots from beyond the 3-point line. This pattern is reflected in the steady increase in the 3-point rate among these positions over time, reaching a point where **approximately 50% of the shots taken by Shooting Guards are now 3-point attempts**.

However, the evolution of the 3-point shot is even more evident when examining the **interior positions**. Historically, Power Forwards (PF) and Centers (C) rarely attempted shots from beyond the arc. For many years, the 3-point rate for Power Forwards remained around **10% of their total field goal attempts**, while Centers rarely exceeded **5%**.

In recent years, this pattern has changed dramatically. Modern offensive strategies have increasingly encouraged interior players to stretch the floor and contribute to perimeter shooting. As a result, by the **2024–25 season**, **41% of the shots taken by Power Forwards were 3-point attempts**, while **Centers reached a rate of 22%**.

This shift illustrates how the modern game has redefined positional roles, with even traditionally interior players adapting to a league-wide emphasis on spacing and perimeter shooting.

6. **Power Forwards & Centers with > 1 3PA per game.**



Focusing on the **interior positions**, it is clear that Power Forwards (PF) and Centers (C) were late adopters of the three-point shot. In the early years, **almost no players in these positions attempted even one three-pointer per game**, reflecting the traditional emphasis on interior scoring.

In the 1990s, **Power Forwards began experimenting with perimeter shooting**, with nearly **20% attempting three-pointers more regularly**, while Centers largely remained focused on inside play, with only **1–2% averaging at least one three-pointer per game**.

Over the last decade, however, the trend has shifted dramatically. By the **2024–25 season**, approximately **90% of Power Forwards** and **45% of Centers** are taking at least one three-pointer per game. This represents a major transformation in positional roles, demonstrating how interior players have increasingly adapted to modern, perimeter-oriented offensive strategies.

Conclusion

The evolution of three-point shooting in the NBA demonstrates a league-wide strategic shift, transforming the three-point shot from a specialized skill for perimeter players into a fundamental offensive tool for every position on the court.

References

Refer to the following GitHub repository to find the data and queries used in SQL for this EDA, Link.

AI (ChatGPT) was used to assist with the refinement of written explanations and the structuring of the exploratory data analysis.

All data collection, data cleaning, SQL queries, visualizations, and analytical insights were developed by the author, with one exception, during the analysis of the TOP 100, could not get the result to get if a player is “active” or “retired”, ChatGPT helped writing that query.